

Any matrix $\xrightarrow[\text{forward/downward pass}]{\text{row op}^n}$ REF $\xrightarrow[\text{backward/upward pass}]{\text{row op}^n}$ RREF

$$\left[\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 0 & 1 & \frac{2}{6} & \frac{11}{6} & -\frac{17}{12} \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{7}{3} & \frac{1}{3} & \frac{20}{6} \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & -\frac{17}{12} \end{array} \right]$$

Defⁿ A matrix is in RREF if ① it is in REF and ② all entries above each pivot are 0.

Claim Every matrix M can be converted to RREF by row opⁿs (Ex RREF(M) is unique)

Pf. First get REF(M). Then kill entries above pivots as follows.

Pick rightmost pivot col say k^{th} with pivot in row i

For each entry, say a in row j above i in col k ,

do $R_j - aR_i$. This kills all entries col k except the pivot,

leaves all cols $< k$ unchanged and hence keeps the

matrix in REF with same pivot positions.

Now work leftward and induct on # of columns

Ex. Would it work if we used a different column order?

$$\begin{array}{ccccc} 1 & a & b & p & x \\ 0 & 0 & 1 & q & y \\ 0 & 0 & 0 & 1 & z \end{array}$$

example $\downarrow$ $R_1 - pR_3$
 $R_2 - qR_3$

$$\begin{array}{ccc|cc} 1 & a & b & 0 & x - pz \\ 0 & 0 & 1 & 0 & y - qz \\ 0 & 0 & 0 & 1 & z \end{array}$$

$$\downarrow R_1 - bR_2$$
$$\begin{array}{ccc|cc} 1 & a & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & \dots \\ 0 & 0 & 0 & 1 & z \end{array}$$

Quiz 1 Mon Aug 24 2pm

Given $Ax=b$, $\text{REF}(A|b)$ help as follows:

$Ax=b$ is consistent $\iff \text{REF}(A|b)$ has no pivot in last column

If consistent sol's of $Ax=b \xleftrightarrow{\text{bijection}}$ (arbitrary value assignments to free variable)

$\text{RREF}(A|b)$ is even better to directly write down all solutions in terms of free vars.

In our example we get

For general $Ax=b$ we do a similar procedure.

$$\left. \begin{aligned} x_1 &= -\frac{7}{3}x_3 - \frac{1}{3}x_4 + \frac{20}{6} \\ x_2 &= -\frac{1}{3}x_3 - \frac{11}{6}x_4 - \frac{17}{12} \\ x_3 &= 1x_3 + 0x_4 + 0 \\ x_4 &= 0x_3 + 1x_4 + 0 \end{aligned} \right\} \xleftrightarrow{\text{Gen sol}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -\frac{7}{3} \\ -\frac{1}{3} \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{1}{3} \\ -\frac{11}{6} \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} \frac{20}{6} \\ -\frac{17}{12} \\ 0 \\ 0 \end{bmatrix}$$

Provisional defⁿ. A subspace of $\mathbb{R}^n$ is a nonempty subset U of $\mathbb{R}^n$ that is closed under addition and scalar multiplication.

Examples. $\{0 \text{ vector}\}, \mathbb{R}^n$ "Observe" Subspaces of.

For any $n \times n$ matrix A , the solution set of $Ax = 0$ is a subspace of $\mathbb{R}^n$.

$\mathbb{R}^2$: $\{0\}, \mathbb{R}^2$, lines thro' 0

$\mathbb{R}^3$: $\{0\}$, of form $\{av \mid a \in \mathbb{R}, \text{fixed nonzero } v\}$
of form $\{av + bw \mid a, b \in \mathbb{R}, \text{fixed } v, w \text{ non proportional}\}$, $\mathbb{R}^3$.